Exact Counting and Overlap Structure of Three Lucas Difference Sequences

Qichao Wang

October 2026

Abstract

Let $L_0 = 2$, $L_1 = 1$, and $L_{n+2} = L_{n+1} + L_n$. We prove that every positive integer represented as $L_i - 5L_j$ with $i, j \geq 1$ has a unique such representation. We characterize the admissible index pairs, count the values generated by bounded indices exactly, and show that the number of positive values at most x is $\frac{1}{2}(\log_\varphi x)^2 + O(\log x)$. We give an exact count using only three boundary blocks and quadratic formulas for the four nonzero residue subsequences. We completely classify the intersection with A191982 as nine exceptional values and two Lucas-multiple families, derive its exact counting function, classify all five repeated representations of A191982, obtain an exact counting function for A191983, and explain the initial zero.

The index region

Write

$$S = \{L_i - 5L_j > 0 : i, j \geq 1\}, \qquad A(x) = \#\{s \in S : s \leq x\}, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

The identities

$$L_{n+5} = 5L_{n+1} + 3L_n \quad (n \geq 0), \qquad L_{j+3} - 5L_j = -2L_{j-2} \quad (j \geq 2) \tag{1}$$

follow from the Lucas recurrence (or by checking two initial values and then applying that recurrence).

Theorem 1. *For $i, j \geq 1$, the difference $L_i - 5L_j$ is positive if and only if*

$$(i, j) = (4, 1) \quad \text{or} \quad i \geq 5 \text{ and } 1 \leq j \leq i - 4.$$

For each $i \geq 5$, these $i - 4$ differences are distinct and lie in

$$3L_{i-5} = L_i - 5L_{i-4} \;\leq\; L_i - 5L_j \;\leq\; L_i - 5. \tag{2}$$

Proof. The Lucas numbers increase strictly from index 1 onward. For $j \geq 2$, the second identity in (1) gives $L_{j+3} < 5L_j$, whereas the first gives $L_{j+4} = 5L_j + 3L_{j-1} > 5L_j$. Monotonicity settles all i for such j. For $j = 1$, one checks $L_3 = 4 < 5$ and $L_4 = 7 > 5$. Equation (2) and distinctness within a fixed i follow from monotonicity of L_j, $j \geq 1$. □

Uniqueness and growth

Theorem 2. *Every $s \in S$ has exactly one representation $s = L_i - 5L_j$ with $i, j \geq 1$.*

Proof. The residues $(L_1, L_2, L_3, L_4) \pmod 5$ are $(1,3,4,2)$ and repeat with period four; in particular, they are distinct. Suppose $L_i - 5L_j = L_k - 5L_\ell > 0$. Reduction modulo 5 forces $i \equiv k \pmod 4$. If $i = k$, strict monotonicity of the Lucas numbers from index 1 gives $j = \ell$.

Otherwise, by symmetry take $k \geq i+4$. Theorem 1 gives $i \geq 4$, $k \geq 8$, and $\ell \leq k-4$. Thus

$$L_k - 5L_\ell \geq 3L_{k-5} \geq 3L_{i-1} > L_i > L_i - 5L_j,$$

a contradiction. Here L_n increases for $n \geq 1$, and $3L_{i-1} > L_i = L_{i-1} + L_{i-2}$ for $i \geq 4$. □

Corollary 3. *For every integer $N \geq 4$, precisely*

$$1 + \sum_{i=5}^{N} (i-4) = 1 + \frac{(N-4)(N-3)}{2}$$

positive values have representations with first index $i \leq N$. Furthermore, the elements of S are nonzero modulo 5, and within each fixed nonzero residue class modulo 5, the blocks of values obtained from consecutive permissible first indices occur in strictly increasing order.

Proof. The count follows from Theorems 1 and 2. Nonzero residues follow from the four-term Lucas residue cycle. Two first indices in the same residue class differ by at least four. If $k \geq i+4$, the proof of Theorem 2 shows every value in the k-block exceeds every value in the i-block. □

Theorem 4. *As $x \to \infty$,*

$$A(x) = \frac{1}{2}(\log_\varphi x)^2 + O(\log x).$$

Consequently, if $b_1 < b_2 < \cdots$ enumerates the positive elements of S, then

$$\log_\varphi b_n = \sqrt{2n} + O(1).$$

Proof. Let $m(x)$ be the largest integer $m \geq 1$ with $L_m \leq x$ (for $x \geq 3$). Binet's formula $L_m = \varphi^m + (-\varphi)^{-m}$ implies $m(x) = \log_\varphi x + O(1)$. Every positive difference with $i \leq m(x)$ is at most x, giving

$$A(x) \;\geq\; 1 + \frac{(m(x)-4)(m(x)-3)}{2}$$

for large x. Conversely, a positive difference with $i \geq 5$ is at least $3L_{i-5}$ by (2), so $i \leq m(x/3)+5$. This gives

$$A(x) \;\leq\; 1 + \frac{(m(x/3)+1)(m(x/3)+2)}{2}$$

for large x. Since both $m(x)$ and $m(x/3)$ equal $\log_\varphi x + O(1)$, these bounds prove the asymptotic. Finally, substituting $x = b_n$ and using the uniqueness of all positive terms gives $n = \frac{1}{2}(\log_\varphi b_n)^2 + O(\log b_n)$; solving for $\log_\varphi b_n$ gives the claimed estimate. □

Exact counting and residue-class unranking

For integral y, write $H(y) = \#\{j \geq 1 : L_j \leq y\}$ and $[z]_+ = \max(z, 0)$.

Theorem 5 (Three-block exact counting)**.** *For every integer $x \geq 11$, let m be the unique integer satisfying $L_m \leq x < L_{m+1}$. Then*

$$\boxed{A(x) = 1 + \frac{(m-4)(m-3)}{2} + \sum_{h=1}^{3} \left[m + h - 4 - H\left(\left\lfloor \frac{L_{m+h} - x - 1}{5} \right\rfloor \right) \right]_+ .} \tag{3}$$

For $0 \leq x < 11$, $A(x)$ counts the members of $\{2,3,6,9\}$ not exceeding x.

Proof. All admissible differences with first index $i \le m$ are smaller than $L_i \le x$. For $i \ge m+4$, (2) gives

$$L_i - 5L_j \ge 3L_{i-5} \ge 3L_{m-1} > L_{m+1} > x.$$

Indeed, $3L_{m-1} - L_{m+1} = L_{m-1} - L_{m-2} > 0$ because $m \ge 5$. The only remaining first indices are $i = m+1, m+2, m+3$. In one such block, $L_i - 5L_j \le x$ is equivalent to $L_j > \lfloor (L_i - x - 1)/5 \rfloor$. There are $i-4$ admissible indices j, so removing the ineligible ones gives the positive-part expression in (3). □

Corollary 6 (Exact ranking and membership)**.** *The published OEIS A191981 has $a(1) = 0$. For $n \ge 2$, its n-th term is the least integer x for which $A(x) \ge n-1$. A finite upper bound is L_N, with $N \ge 4$ the least integer such that $1 + (N-4)(N-3)/2 \ge n-1$. For membership, $x > 0$ determines the first index i mod 4 by $x \equiv L_i \bmod 5$. The corresponding index blocks are disjoint and ordered, so a single Lucas lookup for $(L_i - x)/5$ gives the unique witness if one exists.*

Proof. The first assertion follows from uniqueness and (3). At least $n-1$ positive differences have first index at most N and every such difference is smaller than L_N. For the last assertion, the Lucas residues modulo 5 distinguish i mod 4. If $k \ge i+4$, then $\min B_k \ge 3L_{k-5} \ge 3L_{i-1} > L_i > \max B_i$, so blocks with the same first-index residue occur in increasing order. □

Theorem 7 (Quadratic residue-class enumeration)**.** *Let $S_r = \{s \in S : s \equiv r \pmod 5\}$ for $r = 1,2,3,4$, each listed in increasing order. The blocks of S_r have first indices $i_r + 4k$ ($k \ge 0$), where $(i_1, i_2, i_3, i_4) = (5,4,6,7)$. The cumulative numbers of terms through block k are*

$$F_1(k) = (k+1)(2k+1), \quad F_2(k) = 1 + 2k(k+1),$$
$$F_3(k) = 2(k+1)^2, \quad F_4(k) = (k+1)(2k+3).$$

Set $F_r(-1) = 0$. To obtain the n-th term in S_r, choose the least $k \ge 0$ with $F_r(k) \ge n$, put $t = n - F_r(k-1)$ and $i = i_r + 4k$. Then

$$s_{r,n} = L_i - 5L_{i-3-t},$$

except that $s_{2,1} = 2$ (the isolated $i = 4$ block).

Proof. The residue cycle determines i_r. By Corollary 6, successive blocks are ordered; within a block the values increase as j decreases. The block sizes are respectively $1+4k$, $4k$ for $k \ge 1$ (with one exceptional value for $k = 0$), $2+4k$, and $3+4k$. Summing the lengths gives the four quadratics. The t-th element in a nonexceptional block has $j = i - 3 - t$. Inverting a quadratic inequality determines k exactly. □

The complete intersection with A191982

Let $T = \{5L_k - L_\ell > 0 : k, \ell \ge 1\}$ be the positive part of A191982. Unlike S, the set T can have repeated representations: $2 = 5L_1 - L_2 = 5L_3 - L_6$.

Theorem 8 (Complete intersection classification)**.** *With $E = \{2, 6, 9, 12, 13, 14, 24, 32, 61\}$, one has the disjoint union*

$$\boxed{S \cap T = E \,\dot\cup\, \{3L_n : n \ge 6\} \,\dot\cup\, \{8L_n : n \ge 6\}.} \tag{4}$$

In particular, the only common values outside the two Lucas-multiple families are the nine elements of E.

Proof. Suppose $L_i - 5L_j = 5L_k - L_\ell > 0$. The Lucas residues modulo 5 repeat as $(1,3,4,2)$, so $L_i + L_\ell \equiv 0 \pmod 5$ forces $\ell \equiv i+2 \pmod 4$. By Theorem 1, $i \ge 4$ and $j \le i-4$, except that $(i,j) = (4,1)$ is the first permitted pair. Furthermore, $5L_k > L_\ell$ implies $\ell \le k+3$ for every $k \ge 1$. For $k \ge 2$, the least positive difference in its block is $5L_k - L_{k+3} = 2L_{k-2}$.

Assume first $i \geq 16$. The lower bound $L_i - 5L_j \geq 3L_{i-5} > 5L_{i-7}$ excludes $k \leq i-7$. The bound $5L_k - L_\ell \geq 2L_{k-2} \geq 2L_{i-1} > L_i$ excludes $k \geq i+1$. If $k = i$, the largest permissible ℓ is $i+2$ and $5L_i - L_{i+2} > L_i$. If $k = i-2$, the largest permissible ℓ is $i-2$ and $4L_{i-2} > L_i$. Thus $k = i-d$ with $d \in \{1,3,4,5,6\}$. For $d = 1$, only $\ell = i+2$ can give a value below L_i; all smaller permissible ℓ give $5L_{i-1} - L_\ell > L_i$.

Put $R = L_i + L_\ell - 5L_{i-d} = 5L_j$. The following table gives strict open intervals for R; each endpoint is a multiple of a Lucas number.

d	ℓ	location of R
1	$i+2$	$(5L_{i-5}, 5L_{i-4})$
3	$i-2$	$(5L_{i-7}, 5L_{i-6})$
3	$\ell \leq i-6$	$R < 0$
4	$i-2$	$(5L_{i-5}, 5L_{i-4})$
4	$i-6$	$(5L_{i-6}, 5L_{i-5})$
4	$i-10$	$R = 5L_{i-6}$
4	$\ell \leq i-14$	$(5L_{i-7}, 5L_{i-6})$
5	$i-2$	$R > 5L_{i-4}$
5	$\ell \leq i-6$	$(5L_{i-5}, 5L_{i-4})$
6	$i-6$	$R > 5L_{i-4}$
6	$i-10$	$R = 5L_{i-4}$
6	$\ell \leq i-14$	$(5L_{i-5}, 5L_{i-4})$

All table inequalities hold for $i \geq 16$. To verify them, bound each variable ℓ between its indicated upper endpoint and $L_\ell > 0$. The resulting fixed-shift inequalities are differences of Lucas-linear expressions and hence satisfy $u_{i+2} = u_{i+1} + u_i$. In each case the two initial values $i = 16, 17$ are strictly positive, as checked by direct substitution; the recurrence preserves positivity. The two exact-equality rows follow from the same recurrence and two initial identities. Every interval either misses all admissible $5L_j$ or contains no such value, while $R > 5L_{i-4}$ is impossible since $j \leq i-4$. The only surviving cases are

$$(i,j,k,\ell) = (i, i-6, i-4, i-10), \qquad (i, i-4, i-6, i-10) \qquad (i \geq 16).$$

For the remaining $4 \leq i \leq 15$, the finite possibilities $1 \leq k \leq i-1$, $1 \leq \ell \leq k+3$, and $j \leq i-4$ (with $(4,1)$ included) give:

i	4	5	6	7	8	9	10
$S \cap T$ in the i-block	2	6	13	9, 14, 24	12, 32	61	$\varnothing$

For $11 \leq i \leq 15$ the only two values are $3L_{i-5}$ and $8L_{i-5}$. For all $n \geq 6$ the Lucas recurrence gives

$$3L_n = L_{n+5} - 5L_{n+1} = 5L_{n-1} - L_{n-5},$$
$$8L_n = L_{n+5} - 5L_{n-1} = 5L_{n+1} - L_{n-5}.$$

Thus both displayed families belong to $S \cap T$. They are disjoint from one another by the uniqueness theorem for S, and neither family contains one of the nine exceptional values (the smallest values are $3L_6 = 54$ and $8L_6 = 144$). This proves (4). □

Corollary 9 (Exact intersection count)**.** *For $x \geq 0$, let $I(x) = \#\{t \in S \cap T : t \leq x\}$ and $H(y) = \#\{j \geq 1 : L_j \leq y\}$. Then*

$$\boxed{I(x) = \#\{e \in E : e \leq x\} + \big[H(\lfloor x/3 \rfloor) - 5\big]_+ + \big[H(\lfloor x/8 \rfloor) - 5\big]_+.}$$

Consequently

$$I(x) = 2\log_\varphi x + O(1), \qquad \frac{I(x)}{A(x)} \longrightarrow 0 \quad (x \to \infty).$$

Proof. The three sets in (4) are disjoint and the Lucas numbers increase for $n \geq 1$. The count and the asymptotics follow from $H(y) = \log_\varphi y + O(1)$ and Theorem 4. □

All repeated representations in A191982

Theorem 10 (Five and only five collisions)**.** *The map*

$$(k,\ell) \longmapsto 5L_k - L_\ell > 0, \qquad k,\ell \geq 1,$$

is injective except for exactly five pairs of distinct arguments:

k	p	(ℓ, q)	$5L_k - L_\ell = 5L_p - L_q$
1	2	(1,5)	4
1	3	(2,6)	2
2	5	(4,8)	8
2	6	(1,9)	14
3	4	(2,6)	17

In particular, the set of multiply represented positive values in A191982 is precisely $D = \{2,4,8,14,17\}$*; each has multiplicity two.*

Proof. For $k = 1$ the positive pairs have $1 \leq \ell \leq 3$. For every $k \geq 2$, positivity is equivalent to $1 \leq \ell \leq k+3$, and the values in the k-block lie in

$$[2L_{k-2},\ 5L_k - 1], \qquad 5L_k - L_{k+3} = 2L_{k-2}.$$

Suppose $5L_k - L_\ell = 5L_p - L_q > 0$ with $k < p$. Then

$$L_q - L_\ell = 5(L_p - L_k) > 0, \qquad q \equiv \ell \pmod 4. \tag{5}$$

The congruence follows from the four distinct Lucas residues modulo 5. Hence $q \geq \ell + 4$.

If $k \geq 3$ and $p \geq k+4$, then

$$5L_p - L_q \geq 2L_{p-2} \geq 2L_{k+2} > 5L_k - 1,$$

where $2L_{k+2} - 5L_k = L_{k-1} - L_{k-2} > 0$. This exceeds every value in the k-block. For $k = 1$, $p \geq 5$ is excluded by $2L_3 = 8 > 5L_1 - 1 = 4$. For $k = 2$, $p \geq 7$ is excluded by $2L_5 = 22 > 5L_2 - 1 = 14$; the boundary $p = 6$ gives precisely 14.

It remains to consider $p = k + d$ for $d \in \{1,2,3\}$. For $k \geq 4$ the possibilities in (5) reduce as follows.

If $d = 1$, then $5(L_p - L_k) = 5L_{k-1}$, so $L_{k+2} < 5L_{k-1} < L_{k+4}$ and $q \in \{k+3, k+4\}$. For $q = k+3$ one has $L_\ell = 3L_{k-2} > L_{k-1}$, contradicting $\ell \leq q - 4 = k - 1$. For $q = k+4$ one has $L_\ell = L_{k+2} + 3L_{k-2} > L_k$, contradicting $\ell \leq k$.

If $d = 2$, then $5(L_p - L_k) = 5L_{k+1}$. Since $L_{k+4} < 5L_{k+1}$ and $q \leq p+3$, necessarily $q = k+5$. It follows that $L_\ell = 3L_k > L_{k+1}$, but $\ell \leq k+1$.

If $d = 3$, then $5(L_p - L_k) = 10L_{k+1}$. As $L_{k+5} < 10L_{k+1}$ and $q \leq p+3$, one must have $q = k+6$. Thus

$$L_\ell = L_{k+6} - 10L_{k+1} = 5L_k - 2L_{k+1} =: R_k.$$

For $k \geq 3$,

$$L_{k-2} < R_k < L_{k+2}.$$

But (5) forces $\ell \equiv k+2 \pmod 4$ and $\ell \leq k+2$; either $\ell = k+2$ or $\ell \leq k-2$, a contradiction.

Finally, the cases $k = 1,2,3$ can be checked using (5) and $q \leq p+3$: $k = 1$ yields $(p,\ell,q) = (2,1,5), (3,2,6)$; $k = 2$ yields $(5,4,8)$ in addition to the boundary $(6,1,9)$; $k = 3$ yields $(4,2,6)$. Their values are the five displayed in the table. Since the list exhausts all colliding argument pairs, no value has three representations. □

Corollary 11 (Exact counting for A191982 and A191983)**.** *Let $B(x) = \#\{t \in T : t \le x\}$ and let $U(x)$ count the nonnegative terms of A191983 not exceeding x, for integers $x \ge 0$. Put $b_1 = 3$, $b_k = k+3$ for $k \ge 2$, and let $m(y) = \max\{n \ge 1 : L_n \le y\}$, with $m(y) = 0$ if $y < 1$. Define*

$$K(x) = \max\{3, m(\lfloor x/2 \rfloor) + 2\}, \qquad D(x) = \#\{d \in D : d \le x\}.$$

Then both counts are exact:

$$\boxed{B(x) = \sum_{k=1}^{K(x)} [b_k - H(5L_k - x - 1)]_+ - D(x),} \tag{6}$$

$$\boxed{U(x) = 1 + A(x) + B(x) - I(x).} \tag{7}$$

In particular,

$$B(x) = \tfrac{1}{2}(\log_\varphi x)^2 + O(\log x), \qquad U(x) = (\log_\varphi x)^2 + O(\log x).$$

The n-th term in either sequence can be determined exactly by integer binary search on its counting function.

Proof. In the k-block, the positive candidates are $1 \le \ell \le b_k$ and $5L_k - L_\ell \le x$ is equivalent to $L_\ell > 5L_k - x - 1$. Thus the number of eligible pairs in that block is $[b_k - H(5L_k - x - 1)]_+$. For $k \ge 3$ the smallest positive candidate is $2L_{k-2}$, so all $k > K(x)$ contribute nothing. Theorem 10 removes exactly one extra representation for each $d \in D$ not exceeding x. Since the nonnegative union is $\{0\} \cup S \cup T$, (7) follows from inclusion–exclusion and Corollary 9.

Write $t = \log_\varphi x$. All positive pairs with $k \le m((x+1)/5)$ contribute, while none with $k > m(x/2) + 2$ do; both index bounds equal $t + O(1)$. Summing $b_k = k + O(1)$ gives $B(x) = \frac{1}{2}t^2 + O(t)$. The asymptotic for U follows from Theorem 4 and $I(x) = 2t + O(1)$. □

Remark 12 (The initial zero)**.** *A191981 begins with 0, but 5 never divides a Lucas number: $L_i \equiv 1, 3, 4, 2 \pmod 5$ for $i \ge 1$. Hence $L_i - 5L_j = 0$ has no solution. The Mathematica program linked from A191980 replaces nonpositive differences by zero before taking their union. Thus the published sequence is $\{0\} \cup S$, in increasing order; its zero is a program-generated sentinel, not a Lucas difference. In particular, the number of published terms at most x is $A(x) + 1$ for $x \ge 0$.*

Reference. C. Kimberling, *OEIS A191981: Ordered sequence of nonnegative differences $f - 5g$, where f and g are Lucas numbers*, https://oeis.org/A191981 (accessed October 2026). The related sequences are https://oeis.org/A191982 and https://oeis.org/A191983; the Mathematica code appears under https://oeis.org/A191980.